

Towards a Quantitative Understanding of Salt-Assisted Liquid-Liquid Extraction: Hybrid Modeling of Quaternary Phase Splits for Furfural Recovery

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Abstract

Salt-assisted liquid-liquid extraction (SALLE) is an industrially relevant method for the treatment of hydrothermal liquefaction (HTL) process water, offering a route for recovery of biocrude precursor organics from aqueous waste streams. In this work, a liquid-liquid interaction model was developed for the water-furfural-ethyl acetate system at various sodium chloride concentrations, using furfural as a model component for biocrude due to its abundance in HTL wastewater. A self-contained workflow for quaternary mixture modeling in Julia is presented, introducing a *quasi-ternary* approach that employs the NRTL formulation for interaction parameter estimation. The proposed method accurately predicts phase split predictions for a fixed salt fraction by weight, constituting a benchmark for hybrid first-principles and data-driven modeling of SALLE processes. Experimental tie lines at 2.5 wt% NaCl were obtained and contrasted to model-generated ternary diagrams to validate the quasi-ternary method, while comparisons with literature organic phase compositions for the pure ternary mixture supported cross-validation of the phase split algorithm. Model predictions show strong quantitative agreement with both experimental and literature phase partitioning, both for the pure ternary system and for fixed salt concentrations. Statistical analysis for uncertainty quantification, in combination with the validation methods, further supports our implementation for unsymmetric quaternary mixtures of partially miscible nature.

Keywords: NRTL, Salt-assisted liquid-liquid extraction, furfural recovery, data-driven thermodynamic modeling, ternary phase diagrams.

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1. Introduction

Liquid-liquid equilibrium in a biphasic mixture is achieved when the chemical potential of solutes is equal across the aqueous and organic phases. Solvent selection significantly impacts the efficiency of the liquid-liquid extraction (LLE) process [1]. Macroscopically, solubility differences between aqueous and organic phases, governed by parameters like dielectric constant and density, impact LLE efficiency. Microscopically, specific intermolecular interactions between solvent and solute, such as hydrogen bonding and van der Waals interactions, play a crucial role. For the recovery of organics from aqueous waste streams, moderately polar solvents like chloroform and dichloromethane are effective for extracting certain organic compounds due to complementary polarity and interaction modes, but their toxicity makes them less desirable for commercial use.² Nonpolar solvents with low dielectric constants, on the other hand, exhibit poor extraction capabilities. The efficacy of LLE is limited for aqueous phases which form metastable supersaturated solution of organics in water, with excess organics potentially exceeding known supersaturation points, such as in the process water from hydrothermal liquefaction of biomass (HTL-PW) [2]. This solutropic nature of HTL-PW limits the purity that can be obtained through fractional extraction of a ternary mixture, mimicking azeotropic behavior. Under such circumstances, kosmotropes (order-making) and chaotropes (order-disrupting) can break the pseudo-equilibrium state of HTL-PW [3,4]. Adding a kosmotrope to the LLE of HTC-PW can promote "salting out" and transfer organic solutes to the organic phase. This is also known as salt-assisted liquid-liquid extraction (SALLE). SALLE leverages the thermodynamics of non-ideal mixtures, specifically that the chemical potential of substances is conditioned by the molecular interactions between all pairs of species involved, to effectively separate partially miscible organic-water mixtures using a low-polarity organic solvent [5].

The increasing industrial popularity of SALLE stems from its time- and cost-effective nature. By encouraging greater solute partitioning into the organic phase, salt addition can reduce the volume of extracting solvent needed to achieve a given recovery [6,7]. Because of decreased solvent needs, SALLE has lower projected environmental impact than its LLE counterpart. Another advantage of SALLE over traditional liquid-liquid extraction is the simplicity of its integration with liquid chromatography (LC) techniques for characterization of the extract [8]. Organic extracts obtained by SALLE, cleaner than those recovered via methods of comparable simplicity such as protein precipitation, can be easily processed through LC-MS/MS bioassays, enabling high-throughput automation [9].

Despite being a standard in pharmaceutical separations, particularly in sample preparation for bioanalysis across biological matrices such as blood, plasma, and serum, acquiring a complete molecular-scale understanding of SALLE is challenging [10,11]. While salting-out and salting-in processes have been well documented in research, SALLE performance is highly system-dependent: the wide array of combinations of solvents and salts, all of which differ at a molecular-thermodynamic scale, limits the transferability of observed separation tendencies. As a result, existing studies, including one-factor-at-a-time (OFAT) approaches, provide valuable system-specific insights but do not readily enable broad generalization across extraction systems [8]. The non-uniformity of SALLE systems considered in literature hinders the construction of sufficiently large databases for computational analysis of separation trends. Additionally, the complexity of molecular interactions in SALLE, a multi-species system, adds another layer of difficulty to computational prediction, which further aggravates the database availability issue as researchers

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must rely solely on the limited experimental tie lines available in the literature for the system of interest [12].

In light of these constraints, the development and optimization of new salting-out processes is conditioned by the lack of validated predictive models of quaternary systems of varied miscibility [12]. In the era of forward-thinking technology and iterative process validation, research must meet the need for increased data prediction of the rising techniques in separation sciences.

Activity coefficient models are widely used to describe phase split behavior in partially miscible mixtures [13–16]. In particular, NRTL and UNIQUAC are widely used thermodynamic frameworks for the study of phase equilibrium in biphasic solvent systems [17]. Between these two methods, literature suggests NRTL yields closer fits to experimental data for the specific three-component system consisting of water, furfural, and ethyl acetate [18]. The NRTL formulation relies on excess Gibbs energies to assess phase partitioning, introducing a nonrandomness parameter to represent the non-idealities of a solvent system of low miscibility [19]. First proposed by Chen et al. in 1982, the electrolyte Non-Random Two Liquid (eNRTL) model provides a framework for comprehensive study of electrolyte-containing mixed solvent systems, incorporating mean ionic activity coefficients to account for the attractive and repulsive interactions of electrolytes with hydrophilic and hydrophobic segments, respectively [20,21]. While data-driven models are emerging in the field, as exemplified by recent work introducing a machine learning model for screening of candidate LLE solvents in biobased 2,3-butanediol extraction [22], there is scant data available on many systems to populate such models. As such, models for LLE found in literature often employ first-principles relationships to calculate thermodynamic parameters, often based on NRTL formulations. Recent work by Jo et al. proposes a Python-based NRTL algorithm to regress thermodynamic parameters across different binary mixtures covering 37 biochemical compounds [23]. In this self-contained model, the differential evolution algorithm (DEA) and Nelder-Mead methods were used to regress NRTL parameters, achieving strong agreement with built-in proprietary parameter estimation tools. Neural network models have also been studied for the simulation of liquid-liquid extraction under an NRTL framework [24].

Of particular interest to our research group, hydrothermal liquefaction (HTL) has the potential to fulfill future demand for renewable biofuels obtained from carbonaceous wastes [25]. HTL is a thermochemical process that utilizes the unique properties of water under elevated temperatures (200–370°C) and pressures (4–20 MPa) to convert large biomass molecules into a carbon-rich solid fuel called hydrochar, and a water insoluble liquid biocrude [26,27]. Under HTL conditions, water's dielectric constant, ionic products, and density shift to promote rapid hydrolysis, depolymerization, dehydration, and decarboxylation. As a result of the harsh conditions necessary for HTL, the process produces a carbon-rich aqueous phase that, if properly separated, could enhance the cost-effectiveness of biofuel production by increasing carbon recovery [2]. Among the compounds commonly found in HTL process water (PW), furfural, obtained from cellulose, is a key intermediary to yield valuable organic products such as 2-methyltetrahydrofuran, with applications ranging from small-molecule pharmaceuticals to biofuels [28,29]. Furfural production in HTL is a result of the hydrolysis of cellulose during the formation of sugar monomers, a step that also generates other highly reactive aromatic compounds [30]. Considered within the top 12 value-added products by the US Department of Energy (DOE), furfural has a global production rate of approximately 300,000 tons/year [31]. In recent years, SALLE has been extensively researched for the recovery of furans from HTL PW, with studies

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highlighting its advantages in terms of speed of execution, minimization of organic waste produced, and enrichment factors [32,33].

To address the need for robust computational frameworks in the prediction of salting-out trends, this paper investigates a modeling workflow for the separation of furfural in a water-ethyl acetate system, with relevance to the advancement of HTL as a sustainable energy source via enhanced biocrude recovery. Discrete ternary systems were studied at 0, 1.75, 2.5, 3.75, 5, 7.5, and 10% (m/m) NaCl at varying solvent concentration pairs to uncover the effects of ionic agents on the partitioning of organic solutes in solvent systems of partial miscibility. While previous attempts to calculate ternary mixture phase splits in Python are found in the literature, we could not locate any non-proprietary models specific to the prediction of SALLE outcomes.

While a comprehensive, mechanistic model for SALLE would be of immense value in designing and optimizing processes ranging from biofuels to metal separations and pharmaceuticals, producing such a tool is challenging due to the large non-idealities introduced by salt addition [34]. We propose a self-contained predictive model of SALLE for furfural in a water-ethyl acetate system, using a multistage optimization algorithm for estimation of the NRTL τ parameter matrix. This workflow, developed in Julia, allows for calculation of phase splits at fixed salt amounts, producing representative ternary diagrams that can be easily compared to experimental outcomes. Merging data-driven and first-principles insights, this model serves as a benchmark for future development of self-contained, predictive computational models for biocrude recovery from HTL PW via SALLE.

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2. Methods

To conduct quantitative analysis of biocrude-like phase separation trends, a representative model system was defined. Furfural was chosen as the model component given its large relevance in industrial separations and the availability of experimental studies that report its separation tendencies. To simulate conditions frequently found in SALLE both in industry use and experimental settings, water and ethyl acetate were chosen as the system's low miscibility solvent pair. NaCl was chosen as the salt of interest, due to its high reported efficiency for extraction [33].

The modeling workflow was structured as follows. A NRTL formulation was first implemented to solve the phase equilibrium condition. The τ interaction parameter matrix, which describes binary component interactions and is a necessary input for NRTL modeling, was regressed using a training data set, which consisted of Aspen simulation outcomes at different starting ternary compositions within the target quaternary composition space. Separate τ matrices were obtained for each of the seven salting-out conditions considered and integrated with the NRTL script. Model performance was evaluated against a validation set consisting of unseen compositions within the convex hull defined by training data, and further assessed through comparison with both experimental and literature phase equilibrium data.

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Think about your readers: they are not your professors who gave you an assignment, rather they are genuinely interested in learning from your work (not insuring that they taught it to you).

As such, we need to introduce each set of equations first with the overarching "why" description, followed by the fundamental/theoretical rationale for the step.

2.1 Electrolyte NRTL model formulation

For a component i in a liquid mixture, the NRTL formulation defines γ_i , its activity coefficient, as a function of its interaction with all other components present (represented via summation over the interaction parameters τ), the non-randomness parameter α_{ij} , and the NRTL-

specific weighing factors G_{ij} (Equation 1). τ is defined in terms of the interaction Gibbs free energy between component pairs (Δg_{ij}), ideal gas constant R , and temperature T , and is often obtained via curve fitting to experimental data, yielding fitted parameters A_{ij} and B_{ij} (Equation 2). G depends exclusively on α_{ij} and τ (Equation 3).

$$\ln \gamma_i = \frac{\sum_j \tau_{ji} G_{ji} x_j}{\sum_k G_{ki} x_k} + \sum_j \frac{x_j G_{ij}}{\sum_k G_{kj} x_k} \left(\tau_{ij} - \frac{\sum_k x_k \tau_{kj} G_{ki}}{\sum_k G_{kj} x_k} \right) \quad (1)$$

$$\tau_{ij} = \frac{\Delta g_{ij}}{RT} \approx A_{ij} + \frac{B_{ij}}{T} \quad (2)$$

$$G_{ij} = \exp(-\alpha_{ij} \tau_{ij}) \quad (3)$$

For this analysis, α_{ij} was set to 0.2, consistent with NRTL and eNRTL evaluations of biphasic liquid mixtures [35,36]. A hybrid approach accounted for electrolyte effects, in which the eNRTL formulation was used to generate the synthetic training data set, thereby introducing ion-induced interactions into the simulation workflow. To reduce the number of free variables for parameter estimation, a pseudo-ternary simplification was applied at the solver stage. In this framework, deviations in interactions between the ternary system components are implicitly captured in the regressed NRTL parameters, without the added complexity of incorporating ionic species in phase equilibrium predictions.

2.2 Model training data generation

To generate training data, seven simulations were run in Aspen Plus® V11 using the eNRTL model for physical parameter specifications. These simulations accounted for discrete constant salt concentrations in the system, with a focus on the lower range of salt concentrations by mass percent. Since previous studies report a linear relationship between electrolyte amount and shift in expected solutrope at low salt concentrations, this regime was selected to minimize uncontrolled electrolyte effects that could compromise the validity of the pseudo-ternary approximation [37]. Salt concentration was fixed at 0, 1.75, 2.5, 3.75, 5.0, 7.5, and 10.0 wt% with respect to the total ternary mass. The sensitivity analysis mode was used to collect data points resulting from systematic variations of the feed composition, constrained by a 1:1 to 4:1 organic solvent-to-water ratio known to maximize extraction [10]. Similarly, furfural flow rate was constrained between 1 and 10 kg/hr at 0.5 kg/hr increments, with ethyl acetate defined implicitly through mass balance. Extraction conditions were fixed at 298 K and 1 atm. Sensitivity data sets per salt amount were exported as tables populated with 305 rows, each corresponding to a unique tie line generated from a distinct feed condition. The Aspen-generated data sets were exported into Julia in CSV format for development of a hybrid model of furfural, water, and ethyl acetate phase split at constant salt concentration.

A hybrid approach was chosen for the predictive model, combining data-driven and mechanistic features, as this was found to provide a balance between speed of convergence and reliability in calculating phase splits without overfitting. First, the model was trained on phase split data from Aspen corresponding to the desired salt concentration. With this newly acquired information, a multi-step optimization algorithm was implemented to estimate the τ parameter

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Commented [JLG17]: Do you mean combined ternary of 90% + 10% NaCl? Just not 100% sure I follow

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matrix, initially narrowing down the search to a likely region of convergence (loss ≤ 0.1 for current τ guess) and switching to more powerful but computationally expensive solvers once the broad range of interest was identified. This 3x3 τ matrix was then used for computation of outlet stream compositions upon phase split using a deterministic workflow enforcing the phase equilibrium condition. Informed by quaternary mixture tie lines, this hybrid formulation captures the non-idealities associated with electrolyte presence in a solvent system characterized by immiscibility while employing a quasi-ternary approach, which reduces degrees of freedom to prevent an underdetermined system.

2.3 Regression of τ parameters using simulated data

τ matrices for each set of data were regressed using a three-stage optimization method. The loss function, quantifying cumulative deviation from the phase equilibrium condition across the training points, was defined as:

$$\text{Loss}[g] = \sum_n [\log(\max(y_{1,n}, 10^{-10})) + \log(\max(x_{1,n}^g, 10^{-10})) - \log(\max(y_{2,n}, 10^{-10})) - \log(\max(x_{2,n}^g, 10^{-10}))]^2 \quad (4)$$

The activity coefficients were calculated with the NRTL Gibbs free energy formulation (Equation 1). For each τ guess of the multi-phase optimizer, the associated activity coefficient was computed, and the loss function outcome was calculated for each tie line. The loss at each iteration was averaged and returned to the algorithm to inform the subsequent guess based on the optimality of prior attempts.

The optimization algorithm was constructed with three steps, starting with a global exploration of the six-dimensional τ parameter space using a Sobol low-discrepancy sequence. This generated a high-level map of the loss surface to identify promising regions for more robust optimization, avoiding premature convergence with suboptimal average loss. The solution set was constrained to $[-5, 5]$ for each matrix element, to reduce parameter exploration to the physically meaningful ranges established in literature [38]. 8000 unique sets of τ parameters were evaluated, and the lowest-loss point was returned. A smart objective function was used to refine this guess, calculating loss on a small subset of tie lines selected at random from the training data. A multi-start local search was implemented, selecting 50 neighboring points to the current optimum and performing 20 fine-tuning adjustments for each. The loss was calculated at each iteration, and a threshold of ≤ 0.1 was imposed for optimum selection to prevent overfitting and maintain reasonable runtimes without sacrificing model reliability. Finally, local searches were performed based on parameter sensitivity results, with adaptive step sizes to adjust the matrix elements most likely to minimize loss. Figure 1 summarizes this optimization workflow.

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Commented [JLG23]: I feel like a diagram of all of this might be helpful. What do you think? Is a workflow figure possible?

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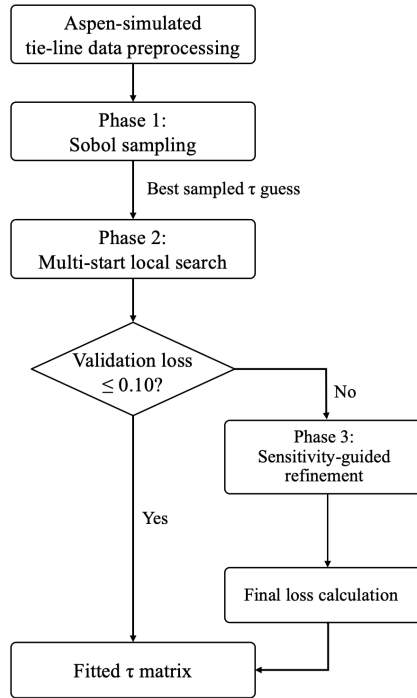


Figure 1. Regression workflow for the obtention of NRTL τ matrices from synthetic tie line data. Candidate parameter sets were first sampled before undergoing local refinement to meet a target loss threshold of ≤ 0.1 .

2.4 Pseudo-ternary liquid-liquid equilibrium (LLE) flash calculations

Flash calculations were implemented using the τ matrix inherited from the optimization workflow. An objective function was established to quantify the deviation from the thermodynamic equilibrium conditions for an initial guess of the compositions of each phase. With the goal of minimizing Gibbs free energy, we defined the objective function as the sum of the squared differences between the activity of each component in the two phases. Thus, for a guess g specifying the phase composition pairs for water, furfural, and ethyl acetate, we have:

$$\text{Err}[g] = \sum_i \left(\gamma_{x,i} x_i - \gamma_{y,i} y_i \right)^2 \quad (5)$$

Where i denotes the components (water, furfural, and ethyl acetate), and x_i and y_i denote the mole fractions of component i in the two liquid phases. A second term was included to capture the error associated with deviations from the constraints derived from the mass balance. By mass

Commented [JLG28]: The way it's written above - water, furfural, and ethyl acetate - could mean that i is any of these three. Yet it's constrained to water and EA being the two phases... so maybe we should be more specific spelling it out?

conservation principles, the feed composition must equal a weighted combination of the two-phase compositions for each component. Extending this analysis to all components in the quasi-ternary formulation ensures that the global mass balance condition is met. The associated penalty function, upon back-calculation of corresponding feed fractions, is:

$$\text{Err}'[g] = 10 \sum_i (z_{i,\text{ref}} - z_{i,\text{calc}})^2 \quad (6)$$

Here, $z_{i,\text{ref}} - z_{i,\text{calc}}$ is the difference between target and calculated compositions of component i in the feed. Note that $z_{i,\text{calc}}$ is a function of the current guess g . A weighting factor of 10 was selected empirically to ensure effective exclusion of unphysical solutions without dominating the overall loss.

Two fallback penalty functions were established to discourage the solver from exploring regions that are of no interest physically, such as the trivial solution in which the compositions of both phases approach the feed condition. The differentiable penalty function p_{trivial} was established to avoid solutions with negligible phase separation:

$$p_{\text{trivial}}[g] = 1000 \exp(-50 \sum_i (x_i - y_i)^2) \quad (7)$$

Similarly, a β penalty function of 10 was applied to phase fractions falling outside the physically meaningful range of 0 to 1. Thus, the overall error was computed as the sum of four deviation quantifiers: phase equilibrium offset, mass balance error, and the two penalty terms enforcing nontrivial and physically plausible solutions. This objective function was wrapped in a phase split solver script to calculate the equilibrium composition of each phase, with a feed composition and a τ matrix as inputs.

2.5 Experimental measurements of equilibrium phase compositions

Experimental liquid–liquid equilibrium data were collected for the water–furfural–ethyl acetate system using a Salt-Assisted Liquid-liquid extraction (SALLE) approach. Six feed compositions were evaluated within a furfural-containing biphasic region selected for tie-line validation. Because the salt-free furfural mass fraction was fixed at 0.05, while the water fraction varied from 0.20 to 0.45, the initial furfural concentration relative to water ranged from approximately 0.111 to 0.249 g mL⁻¹ water, or 1.11×10^5 to 2.49×10^5 mg L⁻¹ water. These values are above the ambient aqueous solubility of furfural, serving as a furfural-rich biphasic validation range. The remaining ternary fraction was ethyl acetate, calculated as

$$x_{EA} = 1 - x_W - x_F, \quad (8)$$

with $x_F = 0.05$.

All assays were prepared gravimetrically in 10 mL Class A graduated glass cylinders. A total ternary mass was selected for each run such that the estimated starting volume was approximately 9 mL. The masses of HPLC-grade water, Furfural (Sigma-Aldrich, ≥ 99.0 mass%),

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Commented [JLG30]: So smart, and so well written!

Commented [SH31]: Can we consider these "low-furfural" when the solubility of furfural is around 0.08 g/mL and the lowest concentration at 0.11 g/mL?

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Commented [JLG33R31]: Good point. Let's not say low. Let's actually just say where it falls in terms of solubility? (also... do we need data below solubility because of this)?

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and ethyl acetate (VWR BDH Chemicals, ≥ 99.5 mass%) were calculated from the target ternary mass fractions according to

$$m_W = x_W m_T, m_F = x_F m_T, m_{EA} = x_{EA} m_T, \quad (9)$$

where m_T is the total ternary mass.

Sodium chloride was added to achieve 2.5 percent by mass in the overall mixture. Defining the overall salt mass fraction as

$$w_s = \frac{m_{NaCl}}{m_T + m_{NaCl}} = 0.025, \quad (10)$$

the NaCl mass was calculated as follows.

$$m_{NaCl} = \frac{0.025}{0.975} m_T \approx 0.0256 m_T \quad (11)$$

The full salt mass was added to the weighed water fraction and mixed until dissolved before the addition of furfural and ethyl acetate.

For experimental planning, component volumes were estimated from the weighed masses using literature densities [39–41]. Water, furfural, and ethyl acetate hypothetical volumes were calculated using $V = m/\rho$, with densities of 0.997 g mL^{-1} , 1.159 g mL^{-1} , and 0.902 g mL^{-1} , respectively. These density-based volume estimates were used only for preparation and not treated as equilibrium measurements. A hypothetical NaCl volume was also included in the preparation table using a constant conversion factor. The total ternary mass was adjusted such that the estimated total starting volume was close to 9 mL. These volume estimates were used only for planning purposes; equilibrium phase volumes were measured directly.

After preparation, each mixture was briefly homogenized using a small magnetic stir bar, capped with Parafilm to minimize evaporation, and placed in a temperature-controlled water bath held at room temperature at approximately 22°C . Samples were equilibrated for 24 hours and then allowed to settle until a stable phase interface was observed. The volumes of the water-rich and ethyl acetate-rich phases were recorded directly from the graduated cylinder.

After phase separation, an aliquot of the water-rich phase was withdrawn using a pipette from the middle of the aqueous layer while avoiding the interfacial region. Aqueous-phase furfural and ethyl acetate concentrations were quantified using a Shimadzu high-performance liquid chromatograph equipped with an Aminex HPX-87H column operated at 35°C . The mobile phase was $4 \text{ mM H}_2\text{SO}_4$ at a flow rate of 0.6 mL min^{-1} , the injection volume was $10 \mu\text{L}$, the analysis time was 60 min, and detection was performed using a UV-vis detector at 210 nm. Samples were diluted with HPLC-grade water as needed before analysis.

To convert HPLC results into model-usable tie lines, a data preprocessing step was performed. First, the triplicate raw readings for the water-rich phase (mg/L) were averaged arithmetically after confirming that no outliers were detected via Grubbs and Dixon Q tests, with $\alpha = 0.05$. Next, the masses of furfural and ethyl acetate in the water-rich phase were obtained as the product of the mean HPLC concentration excluding outliers and the initial dispensed water volume, under the assumption of negligible volume shrinkage from ternary mixing. The remaining amount of water was found by subtraction from the starting water mass, neglecting the effect of solutes in the water-phase density and assuming retention of NaCl in the aqueous phase. Organic

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phase compositions were obtained by mass balance using feed compositions and measured values for the aqueous phase. Error introduced by these simplifications was minimized by experimental design, as low furfural concentration was maintained across all experiments. The resulting composition vectors were then normalized to mass fractions excluding salt to constitute the experimental tie line set.

The OP samples are analyzed on a Shimadzu GC-2010 Plus Gas Chromatograph-Mass Spectrometer (GC-MS) with AOC-20i autosampler on a Restek Rtx-5Sil MS (crossbond 30 m long 0.25 mm ID) fused silica column. The GC-MS samples are prepared by syringe-filtering 1 mL of OP into 1.5 mL GC-MS vials, using a 0.45 μ m hydrophilic PTFE filter. The GC-MS oven temperature is set at 40 °C for 5 min, then ramped at 5 °C/min to 300 °C where it is held for 5 min before ramping down to 250 °C at 20 °C/min, where data collection stopped. Interface and ion source temperatures were 320 and 230 °C, respectively. The sample is injected at 280 °C with a flow of 1 mL/min helium and a split ratio of 10:1. After the solvent cut time of 5 minute, the mass spectrometer scans with the event time (scan period) of 0.4 s from 15 to 400 m/z in electron ionization mode. Each GC-MS run includes the sampler performing five pre-run solvent rinses of the injector needle, three rinses with sample, and five post-run solvent rinses. Compounds are identified by spectra through the internal NIST library (match >70%) and are quantified using an in-house calibration of 88 standard biofuel compounds. Specifically for furfural, the calibration points span from 5 mg/L to 100 mg/L. The open-source Python package *gcms-data-analysis* [42] was used to analyze the data.

2.6 Model validation

To perform a thorough assessment of our computational methodology, both in the baseline case and under non-idealities from salt addition, a series of validation steps were performed to test the reach and accuracy of the model. Initially, model performance was assessed against the original training data set containing synthetic tie lines generated in Aspen Plus® for the pure ternary system. To evaluate the accuracy of the quasi-ternary model at predicting the deviations originating from NaCl addition, validation runs were performed across six synthetic datasets gathered at fixed salt mass fractions. For this purpose, composition simplexes were chosen from within the convex hull defined by the training data, with exact matches with training tie lines excluded from the validation set to ensure that purely predictive capabilities are tested. To further validate the proposed hybrid workflow for the prediction of phase split behavior inside the binodal region, NaCl concentration was fixed at 2.5% (m/m) following *in silico* observations, to capture significant electrolyte-induced deviations in solubility patterns while maintaining a sufficiently low salt concentration to remain within the scope of the pseudo-ternary assumption. The model was then tested against the experimental tie lines derived from HPLC measurements by running the solver on the experimental feed compositions. Trends shown by the Aspen training set were displayed for reference by mapping to the nearest Aspen tie lines in terms of feed composition; feeds within Euclidean distance $d < 0.005$ in the ternary composition space from an existing tie line in the Aspen data set were directly compared, while the others were assigned a weighted interpolation between the two neighboring Aspen tie lines. A final validation step involved comparison with literature equilibrium organic phase compositions from the data set gathered by de Almeida et al., for which feed compositions in the trajectory of the literature tie lines were randomly generated and used for model predictions.

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3. Results and Discussion

This work aimed to develop a predictive model for phase splitting in the SALLE extraction of furfural by integrating a first-principles NRTL framework with synthetic, literature, and newly collected experimental data. Model building was followed by a sequential validation scheme in which simulated, literature, and new experimental data sets were used. A baseline validation was performed using synthetic tie lines at varying salt concentrations, while published tie line data and new experimental runs were used to further assess prediction accuracy and model robustness.

3.1 τ fitting

Table 1 shows the set of τ parameter guesses that resulted from performing the optimization routine on the training set containing Aspen simulated data points, for each of the electrolyte concentrations studied.

Table 1

Regressed τ Parameters from Synthetic Tie Line Data Set Across Salt Concentrations

NaCl Conc. (%m/m)	τ Matrix Elements								
	τ_{11}	τ_{12}	τ_{13}	τ_{21}	τ_{22}	τ_{23}	τ_{31}	τ_{32}	τ_{33}
0.00	0.000	4.467	4.741	2.840	0.000	3.957	2.751	0.810	0.000
1.75	0.000	4.123	4.310	2.052	0.000	-0.552	2.859	0.898	0.000
2.50	0.000	3.004	4.030	2.312	0.000	-1.877	2.844	-0.322	0.000
3.75	0.000	3.198	4.056	0.346	0.000	1.293	3.279	0.424	0.000
5.00	0.000	3.001	4.052	2.536	0.000	0.424	3.034	0.295	0.000
7.50	0.000	2.134	3.750	2.552	0.000	-1.615	3.172	-0.419	0.000
10.00	0.000	4.030	3.921	2.473	0.000	2.129	3.340	1.970	0.000

Note: NRTL τ parameters, quantifying interaction strength between species in the ternary mixture containing water, ethyl acetate, and furfural at varying NaCl concentration, were obtained following a multi-step regression workflow. A τ matrix was fit to Aspen-generated training data for each salt concentration, containing 305 reference tie lines modeled under the eNRTL formulation. Feed compositions in the training set spanned 1-11 wt% furfural, 20-56 wt% water, and 33-79 wt% ethyl acetate.

The associated root-mean-squared errors ($RMSE_{logK}$) for evaluating thermodynamic consistency in the fitting step are provided in Table 2. To correct for potential misrepresentation of prediction error contributions from the more dilute components of each phase, a weighted $RMSE_{logK}$ is also shown, where RMSE is normalized by the average composition across the two existing phases [43]. The exact weighing algorithm used can be found in the Supplemental Information. The justification for reporting this metric lies in the practical effects of quantifying model deviations from the reference phase equilibria behavior: the overall root-mean-squared error, being a result of additive measures, will have greater contributions from the compounds that exist in larger quantities in the overall mixture. For practical applications, maximal recovery of the organic solute (such as furfural) is often the goal. However, the organic solute is most often the quantity present in the smallest proportion. This is frequently the case with HTL post-processing, as it is common to encounter in literature the treatment of PW with 10-20% biocrude content [44,45]. Because this study defines compositions relative to the total ternary mass, the above range

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corresponds to a furfural mass fraction upper bound of 0.05-0.1, with each solvent's content approaching five to ten times this amount. As a result, the underrepresentation of significant deviations in furfural content predictions can be misleading if the cumulative unweighted $RMSE_{logK}$ is used as the sole metric for model fitting assessment.

Table 2
Statistical Metrics for Assessment of Model Performance in τ Parameter Regression

Salt Conc. (%m/m)	Overall Metrics		Component-wise Metrics					
			Water		Furfural		Ethyl Acetate	
	RMSE log K, uw	RMSE log K, w	RMSE log K, uw	RMSE log K, w	RMSE log K, uw	RMSE log K, w	RMSE log K, uw	RMSE log K, w
0.00	0.048	0.042	0.015	0.015	0.050	0.053	0.064	0.062
1.75	0.082	0.056	0.039	0.039	0.012	0.012	0.066	0.067
2.50	0.138	0.095	0.048	0.048	0.198	0.171	0.126	0.127
3.75	0.165	0.136	0.106	0.106	0.213	0.240	0.158	0.156
5.00	0.134	0.087	0.042	0.042	0.202	0.215	0.108	0.108
7.50	0.147	0.091	0.060	0.060	0.225	0.225	0.101	0.102
10.00	0.123	0.111	0.080	0.080	0.015	0.014	0.014	0.014

Note: $RMSE_{logK}$ values are provided to quantitatively assess the τ parameter fitting step, both for the overall mixture and in a per-component basis. Both unweighted (uw) and weighted (w) metrics are provided, with a weighing algorithm used to quantify deviations relative to the scale of each composition, allowing for direct recognition of potential biases in the global RMSE under highly dilute conditions.

As evidenced qualitatively, no direct trends are found from the fitting of the individual τ parameters themselves, which is to be expected due to the data-driven (as opposed to deterministic) approach that was taken for matrix fitting. Element-wise τ comparison plots are included in the Supplemental Information. Since optimization modifies all parameters simultaneously, it is inherent to this methodology for the composite guess to carry greater practical significance than the element-wise analytical counterpart.

3.2 Model validation

3.2.1 Baseline validation of the ternary model (water-furfural-ethyl acetate)

To assess the predictive capabilities of the model, we performed a baseline validation workflow on the pure ternary system using synthetic data. For this purpose, feed compositions not represented in the τ regression training set were generated by linear interpolation between Aspen-generated tie lines within the two-phase region. This out-of-sample validation strategy provided in-domain results showing first indications of generalization in our proposed modeling workflow, for the simplified case of no salt present during extraction. Figure 2 shows the simulated phase split compositions, with associated feeds obtained from interpolation by random sampling of adjacent feeds in training data. Aspen tie line trends are displayed for comparison.

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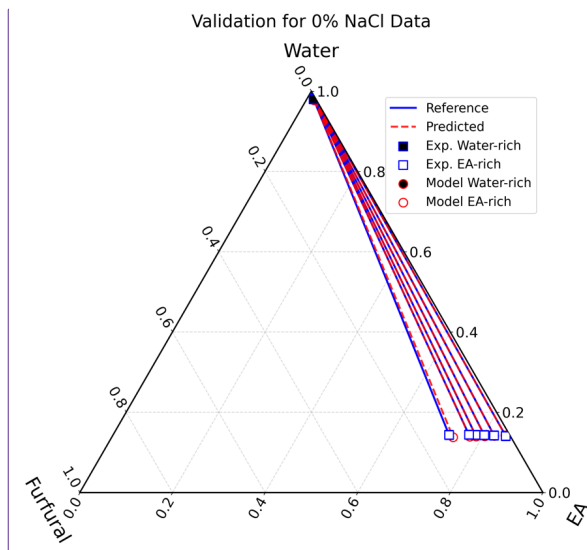


Figure 2. Ternary diagram of a water, furfural, and ethyl acetate-containing ternary system exhibiting no salt addition. Model-generated tie lines for feed conditions generated by random sampling and subsequent interpolation of adjacent training tie lines are displayed in terms of molar fractions. Evenly spaced, representative tie lines from the synthetic training data set are shown as reference of expected binodal region boundaries. RMSE and R^2 are included for fit quality assessment.

Inspection of the ternary plot suggests strong agreement between model outputs to interpolated feeds from the training data set and the corresponding interpolated tie line composition. A RMSE of 0.003, well below 0.01, is a strong indicator of near-absolute model-reference data correspondence. We note that, while reported for consistency, R^2 tends to overrepresent the quality of the pure ternary fit due to unperturbed linearity in solubility trends, which results in the interpolation feed set not truly constituting an out-of-sample data set. Consequently, RMSE should be considered as a more physically significant metric for quantifying model agreement with synthetic data for the pure ternary mixture. Overall, this baseline validation constitutes a verification workflow for the NRTL routine and the optimization algorithm.

3.2.2 Validation of the quasi-ternary approach (water-furfural-ethyl acetate + salt)

Validation performed on interpolated tie lines showed strong predictive performance upon visual assessment of phase split patterns, as shown in ternary diagram form in Figure 3. Diagrams (A), (B), (C) and (F) present instances of unphysical separation predictions, which appears as a potential instance of model sensitivity to poor initial guesses. No unphysical tie lines are produced for (D) and (F), as reflected in the corresponding R^2 values.

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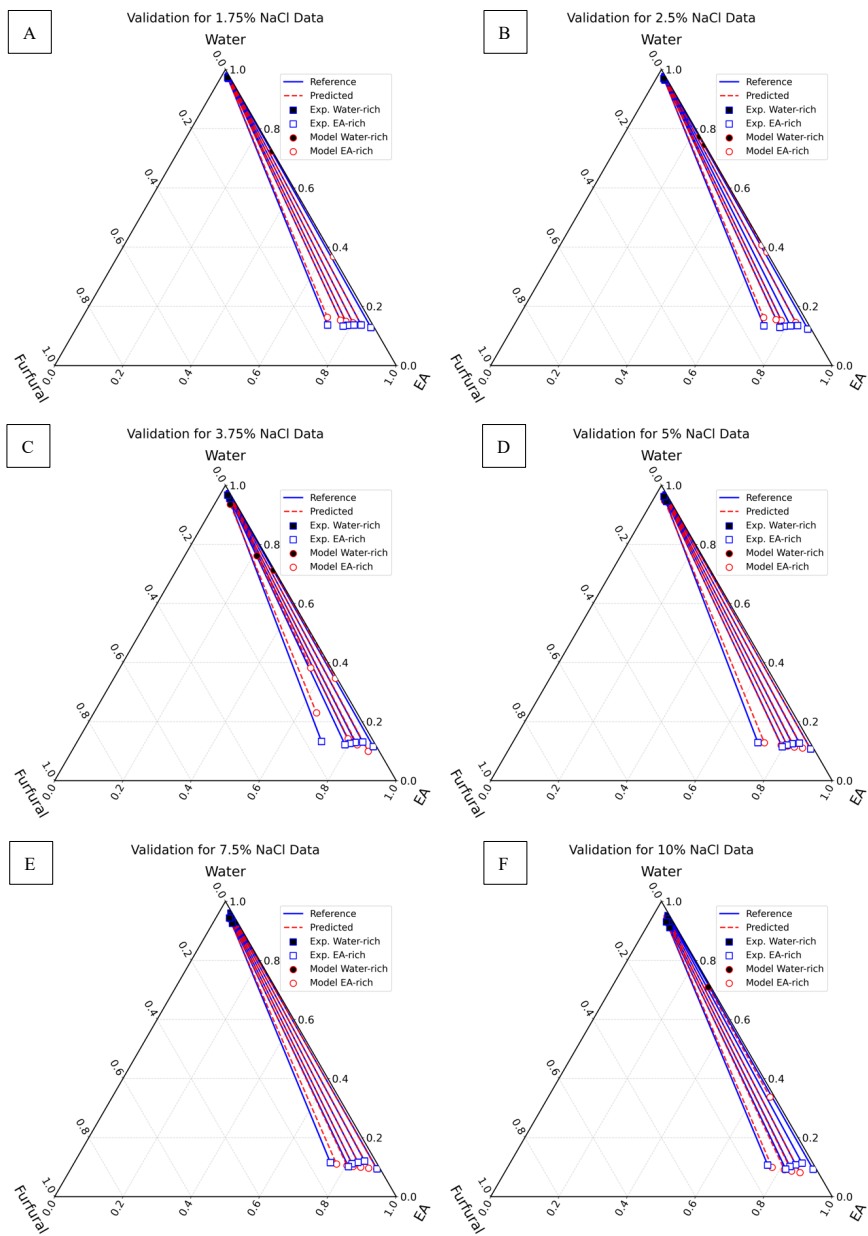


Figure 3 (A-F). Ternary diagrams showing model-predicted phase split tendencies in comparison to reference tie lines, which were generated by linear interpolation between adjacent training feed points and by averaging the corresponding synthetic tie lines. Tie lines for systems of fixed sodium chloride concentrations representing 1.75, 2.5, 3.75, 5, 7.5, and 10% (m/m) with respect to the total quaternary mixture are shown on a mole basis. A quasi-ternary approach, under the assumption of approximately constant salt composition throughout the separation process, was used to reduce the quaternary system to a three-dimensional space, a well-characterized method in literature [46]. RMSE and R^2 are included for fit quality assessment.

Table 3 shows the calculated RMSE and R^2 for validation on interpolated tie lines from the synthetic training set, for the salt concentrations studied (including the pure ternary case). The mean RMSE and R^2 across all salt concentrations were 0.058 and 0.965, respectively.

Table 3

Statistical Metrics for Validation on Synthetic Intermediate Feeds Using Interpolated Tie Line Endpoints

Salt Conc. (%m/m)	Overall Metrics		Component-wise Metrics		
	RMSE	R^2	Water	Furfural	Ethyl Acetate
0.00	0.003	0.999	0.003	0.002	0.004
1.75	0.081	0.959	0.100	0.005	0.098
2.50	0.110	0.923	0.139	0.016	0.130
3.75	0.110	0.923	0.139	0.014	0.130
5.00	0.009	0.999	0.010	0.006	0.011
7.50	0.011	0.999	0.013	0.005	0.013
10.00	0.085	0.952	0.104	0.004	0.104

Note: This model assessment step was performed by generating a validation set within the region bounded by the composition triplets used for training. The validation set was constructed by selecting six tie lines evenly spaced throughout the data set sorted by incremental composition changes to represent the training space. The feed compositions associated with these tie lines were averaged with the immediately neighboring feed point, in the interest of constructing a set of unseen data. Similarly, corresponding synthetic tie line compositions were interpolated on a per-component basis for comparison with model-predicted trends.

3.2.3 Model validation against experimental tie lines

The ternary diagram contrasting experimentally obtained and model-simulated tie lines is presented in Figure 4. Statistical parameters for quantitative fit assessment are reported in Table 4.

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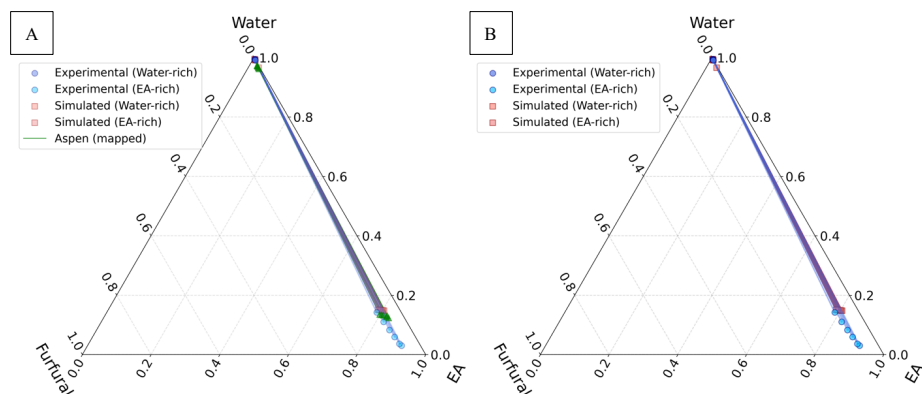


Figure 4 (A-B). Comparison between simulated tie lines and experimental binodal curve points for a feed composition subset within the model training region, in molar fractions, with (A) and without (B) the corresponding Aspen eNRTL predictions. NaCl amount was fixed at 2.5% m/m. Experimental compositions in the water phase at equilibrium were determined by measuring furfural and ethyl acetate concentrations in the aqueous phase using HPLC. The organic phase compositions were estimated by mass balance and further corroborated with GC-MS measurements of furfural concentration in the ethyl acetate-rich extract.

Table 4

Statistical Metrics for Validation on Experimental Tie Lines

Salt Conc. (%m/m)	Overall Metrics			Water-rich Phase			Ethyl Acetate-rich Phase		
	RMSE	MAE	Bias	RMSE	MAE	Bias	RMSE	MAE	Bias
2.5	0.006	0.006	-0.002	0.003	0.003	+0.003	0.008	0.008	-0.008

Note: HPLC data, with an average relative uncertainty of 3.97%, were collected and used for the calculation of the equilibrium mass fractions of furfural and ethyl acetate in the aqueous phase. Root-mean-squared error (RMSE), mean absolute error (MAE) and bias in the prediction of compositions per phase are reported.

From Figure 4, it appears that water content in the organic phase is systematically overpredicted by the model. A possible explanation lies in how water composition is estimated in our experimental setup: from HPLC data, measurements for furfural and ethyl acetate are obtained for the aqueous phase, and water composition is obtained by subtraction from the initial water volume added. However, the solubility of water in ethyl acetate is non-negligible, even within the binodal region. Literature tie lines report compositions of 18-25 mol% water in the organic phase for water-to-ethyl acetate (W:EA) ratios ranging from 4:1 to 16:1 [18]. While the highest ratio tested experimentally was 0.9:1 W:EA, it is possible that the assumption that the initial dispensed volume of water can be used to estimate the water composition is not valid in this experimental regime, though further analysis should be performed to reliably conclude this. Even with suboptimal water composition predictions, model-produced furfural compositions closely align with experimental observation, which suggests that the model reliably predicts solute recovery in the tested composition space. Low RMSE and MAE parameters overall show close overall agreement between model and experimental separation trends. Ethyl acetate dominates bias in the organic phase because it is most abundant, and apparent water overprediction corresponds to apparent ethyl acetate underprediction. A comparatively large, negative bias in the organic phase relative to the aqueous phase is consistent with an apparent

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underprediction of ethyl acetate relative to water. However, given the limitations of the assumption used to determine the experimental tie lines, it is not clear whether these results reflect a true mismatch between model and experimental observations. Validation against literature will be performed to further assess the reliability of model predictions beyond furfural recovery.

GC-MS analysis was used to provide independent measures for the composition of the ethyl acetate-rich phase to assess the validity of the mass balance closure assumption. However, organic phase GC-MS compositions were found inconsistent with mass balance estimates in furfural content, up to one order of magnitude (GC-MS: <0.34-0.49 mol% furfural; mass balance: 5.54-7.16 mol% furfural). This discrepancy is partially attributed to gravitational settling of furfural ($\rho=1.159$ g/mL) as a sub-layer in the ethyl acetate phase prior to sampling from the mixture vial, given the large density difference with ethyl acetate ($\rho=0.902$ g/mL), which was observed visually in the ternary mixture. Another consideration is that expected furfural concentrations in the organic phase exceeded the GC-MS calibration range, rendering quantification unreliable. Furthermore, since water is not accurately detected in GC-MS under organic-rich conditions, ternary composition reconstruction from GC-MS readings is limited. GC-MS data were thus excluded from model validation steps. It was noted that organic phase compositions estimated from mass balance are subject to error propagation from HPLC measurements, as no independent organic phase measurements were incorporated.

3.2.4 Model validation against literature

In the interest of assessing the accuracy of the model in predicting binodal points, cloud point data reported by de Almeida et al. for the pure ternary system (LLE) [18] was compared to tie line endpoint predictions at the organic-phase branch. In the de Almeida work, the binodal curve for a furfural, water, and ethyl acetate ternary system was reconstructed by experimental measurement of cloud points, composition triplets at which phase split begins to be observed, at different temperatures. Cloud points for the organic phase were determined by adding water to a furfural and ethyl acetate mixture until a faint turbidity was observed. The literature compositions in the ethyl acetate phase following salting-out extraction using the cloud point method are reported in Table 5. The comparison of predicted tie lines to organic phase compositions reported by de Almeida et al. is shown in Figure 5.

Table 5

Cloud Points for Water + Furfural + Ethyl Acetate Mixture – Organic Phase at 303.2 K as Reported by de Almeida et al. [18]

Tie line #	Ethyl acetate-rich phase composition		
	W _{water}	W _{furfural}	W _{ethyl acetate}
1	0.0346	0.0000	0.9654
2	0.0404	0.1157	0.8439
3	0.0480	0.2378	0.7142
4	0.0507	0.3586	0.5907
5	0.0566	0.4729	0.4705
6	0.0574	0.5716	0.3710
7	0.0603	0.7538	0.1859

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8	0.0602	0.8052	0.1346
9	0.0593	0.9407	0.0000

Note: These data correspond to Table 4 in the de Almeida work.

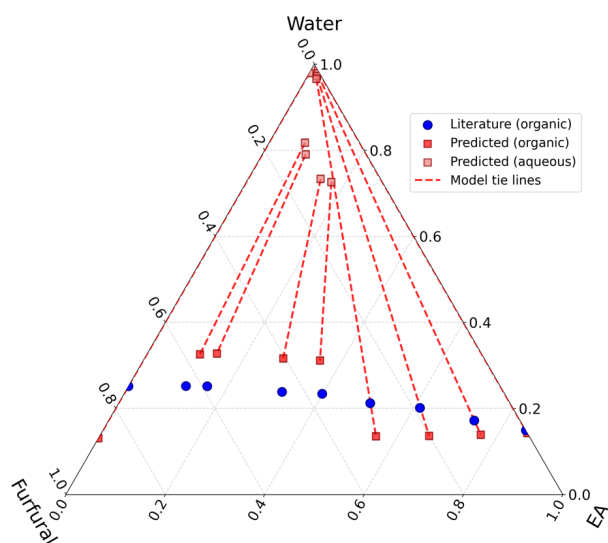


Figure 5. Ternary diagram showing model-predicted tie lines and organic phase compositions for the furfural, water, and ethyl acetate mixture as reported by de Almeida et al. [18], presented on a mole basis. The legacy data set for $T=303.2$ K was used as it was closest to the conditions used for τ regression and NRTL-based phase equilibria calculations ($T=298$ K). Feed compositions in the pure water to the literature organic composition trajectories were used, consistent with the phase separation behavior observed in the de Almeida study and our experimental validation.

A noticeable pattern is that furfural concentrations in the organic phase appear to be better predicted for points corresponding to a tie line extending within the low-furfural region. Similarly, water phase predictions weaken significantly for tie lines with possible feed points extending beyond the low furfural region (<0.1) of the diagram. This is an expected observation, given our training sample, which focused exclusively on the low furfural region, as the broader purpose of this study is to generate robust predictions within the range of interest for solute recovery from HTL PW. While larger levels of organic solutes are seen in other applications such as the processing of raw biomass [47], these implementations are beyond the scope of this paper. However, while validation on a high-solute region is not performed in this work, our modeling workflow could be evaluated on such a data set by using known representative tie lines on the training step, whether synthetic or experimental.

For further assessment of predictive capabilities, parity plots were produced contrasting model-generated and literature data for water, furfural, and ethyl acetate compositions in the organic phase (Figure 6).

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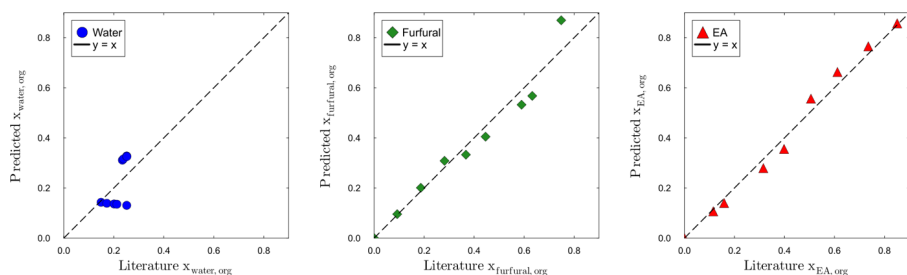


Figure 6. Parity plots for comparison of model-predicted ($T=298\text{ K}$) and literature (de Almeida et al. [18], $T=303.2\text{ K}$) compositions in the ethyl acetate-rich phase, in a pure ternary system containing water, furfural, and ethyl acetate. Molar fractions of water (left), furfural (center), and ethyl acetate (right) in the organic phase are represented.

Water presence in the organic phase appears systematically underpredicted for the experimentally lower water molar fractions, which correspond to a feed condition with larger water content as estimated in our simulation workflow. Interestingly, model tendencies shift toward overprediction of water content in the organic phase for the equilibria arising from lower water fractions in the feed. As observed in Figure 5, model-generated tie lines in the low ethyl acetate region show significant prediction error not only for organic phase composition in literature, but also for expected aqueous phase compositions. Given both (i) the discontinuity in water-side endpoints starting at tie lines crossing feed compositions with $x_{EA} \approx 0.5$ and (ii) the documented trends in aqueous phase composition in our experimental workflow as outlined in section 3.2.3, we expect aqueous phase compositions to consistently cluster near the water apex. This intuition is consistent with the ternary diagrams reported alongside the legacy data set [18]. Discontinuities in predictive trends can be rationalized by considering that model training was performed on synthetic data in the low-furfural, moderate-ethyl acetate range, representative of most extraction instances in furfural recovery process development in the literature, which feature a solvent-to-feed ratio greater than or equal to 1:1 [48–51]. Therefore, inconsistent model predictions are expected for low ethyl acetate and high furfural feeds and thus should not be taken as an accurate assessment of prediction quality.

The observable continuity in predictions corresponding to the upper half of the ethyl acetate axis, both for aqueous and organic phases, suggests that the model is well-behaved within the scope of the training set. Furfural and ethyl acetate parity plots show no systematic offsets, with balanced scattering around the diagonal and tightly clustered observations in the low-composition regions, which suggests that prediction error scales positively with an increase in molar fraction for furfural. Ethyl acetate compositions in the organic phase are best predicted for low-content and high-content ethyl acetate feeds, with the largest deviations occurring at intermediate compositions. With ethyl acetate widely regarded as one of the highest-ranking candidate solvents for furfural extraction procedures [52], we can rule out the possibility of upper bounds in furfural solubility affecting practically achievable concentrations in experiments and resulting in overprediction at high furfural molar fractions. One potential cause for these discrepancies is the temperature gap between literature and training tie lines of 5.2°C . De Almeida [18] observes a noticeable shift in tie line endpoints in the 288.2 K to 303.2 K temperature range, which suggests that even a 5-degree temperature offset between reference data and model setup might be significant in quantitative evaluation.

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In the interest of investigating whether the effects of increasing temperature in a water, furfural, and ethyl acetate mixture mimic salting-out effects at unchanged temperature, model predictions using the τ matrix computed for the 2.5 wt% sodium chloride training set were also fitted to the literature tie lines (Figure 7), and a parity plot was generated (Figure 8). As seen in these figures, model-predicted organic compositions show substantial agreement with the reference values. An analysis of published LLE tie lines for furans in ternary systems involving an organic solvent and water shows that the organic phase tends to exhibit higher water content at higher temperatures [53,54], a tendency that can be explained by the slight increase in the miscibility of water and organic solvents such as ethyl acetate [55]. It has been noted in literature that for some solvent system and salt pairs, such as hydrophobic deep eutectic solvents (HDES) coupled with a water-insoluble ammonium salt, the salting-out effects not only favor the redistribution of furfural into the organic phase but also increase the water uptake into the hydrophobic phase [56]. However, this does not appear to be the case for furfural extraction under a water-ethyl acetate solvent system, in agreement with the intuitive notion that increasing the overall polarity of the water phase by salt addition further expands the miscibility gap [57]. A more likely possibility is that the quantity of training data for the pure ternary model, in combination with the potential bias introduced by using fully first-principles data without the thermodynamic non-idealities from salt addition in training, introduces hypersensitivity to experimental deviations, thus failing to identify the true (experimental) equilibrium state in favor of small perturbations to reach a local minimum. While relaxing equilibrium conditions would decrease the solver's robustness and is thus not advised, we expect that training the model on a combination of experimental and synthetic phase split data eliminates this concern, if such data were available.

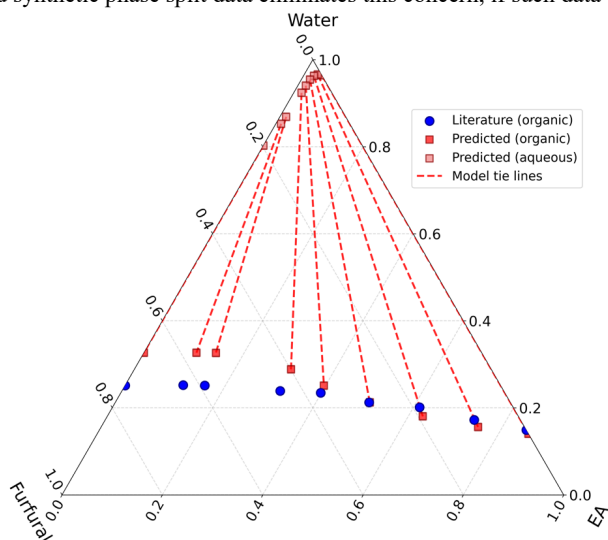


Figure 7. Ternary diagram showing organic phase compositions for the furfural, water, and ethyl acetate mixture as reported by de Almeida et al. [18], presented on a mole basis. Model-predicted tie lines for 2.5% (m/m) salt are overlaid, for evaluation of potential counteracting effects of the non-idealities introduced by salt addition in Aspen-inherited biases. The literature data set for $T=303.2$ K was used as it was closest to the conditions used for τ regression and NRTL-based phase equilibria calculations ($T=298$ K). Feed compositions pertaining to the trajectory from pure

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water to literature organic compositions were used, which is consistent with the phase separation behavior observed in the de Almeida study and our experimental validation.

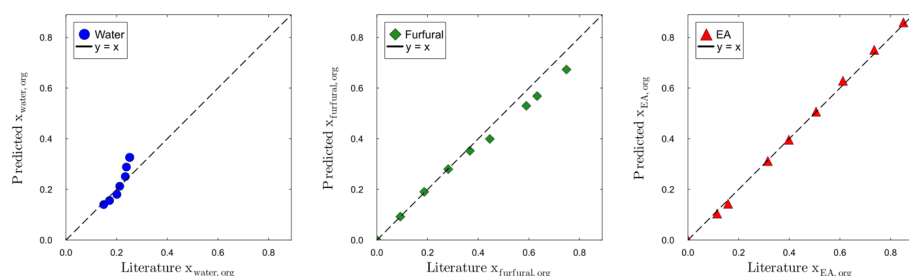


Figure 8. Parity plots for comparison of model-predicted ($T=298$ K, 2.5% m/m) and literature (de Almeida et al. [18], $T=303.2$ K, no salt addition) compositions in the ethyl acetate-rich phase, for a system containing water, furfural, and ethyl acetate. Molar fractions of water (left), furfural (center), and ethyl acetate (right) in the organic phase are represented.

3.3 Uncertainty propagation

Model sensitivity to perturbations in τ matrix elements during NRTL fitting was statistically quantified using the Morris method for the 2.5% (m/m) model. For reliability in ranking factors in order of importance in the overall phase splitting outcome, μ^* , a variation of the μ parameter proposed by Morris in 1991 [58], is used. Eliminating the need for variance-based sensitivity measures, μ^* , (which is computed using absolute values to avoid misrepresentation of negative contributions in non-monotonic systems) is the current standard metric for parameter importance ranking [59]. We found that phase split predictions were most sensitive to changes in matrix entries τ_{13} and τ_{31} , corresponding to interactions between water and ethyl acetate, larger than all other effects by a magnitude of 10^3 - 10^4 x. This is a remarkable observation, as it may suggest that the nature of interactions between the organic solute and solvent (in our case, furfural and ethyl acetate) has minor influences on phase partitioning predictions within the concentration region evaluated in this work. Under the premise that these sensitivity outcomes are not only a measure of model robustness, but also a representation of the real physical contribution of solute-solvent interactions in the behavior of salting-out systems, this result may be an indicator of potential applications of this direct workflow to more complex organic solutes. A necessary step in testing this hypothesis would involve the modeling of two organic solutes amounting to the same concentration as furfural in our salting-out experimental runs. Under the assumption that solute-solute interactions may be approximated as single-component interactions (and thus modeled as such for NRTL calculations), this quantitative workflow shows promise for scaling up to more complex salting-out systems, with the long-term goal of modeling multi-component biocrude solute extraction.

Table 6 shows the summarized results from uncertainty propagation. A detailed report of Morris method outcomes is provided in the Supplemental Information.

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Table 6*Uncertainty Propagation Outcomes in a 2.5 wt% Salt Quasi-Ternary System (Water + Furfural + Ethyl Acetate)*

Feed composition (mol%)			Equilibrium Metrics							
Water	Furfural	Ethyl acetate	Water		Furfural		Ethyl acetate		Log K	
			Mean (aq)	σ (aq)	Mean (org)	σ (org)	Mean (org)	σ (org)	Mean	σ
0.205	0.110	0.685	0.969	-	0.117	-	0.730	-	2.57	-
0.302	0.120	0.578	0.965	1.02e-3	0.142	1.07e-3	0.690	5.34e-3	2.51	1.33e-2
0.410	0.120	0.470	0.964	1.32e-3	0.165	1.27e-3	0.662	5.29e-3	2.48	2.09e-2
0.513	0.103	0.385	0.964	1.27e-3	0.170	1.17e-3	0.659	4.62e-3	2.48	1.77e-2
0.610	0.103	0.287	0.962	1.13e-3	0.206	1.52e-3	0.610	4.47e-3	2.43	1.66e-2

Note: Uncertainty propagation from the τ regression step into phase split predictions was assessed by generating 20 instances per feed of random multiplicative perturbations bounded by a $\pm 2\%$ margin to matrix elements τ_{13} and τ_{31} , identified by the Morris method as most influential to simulation outcomes. Outliers were removed from the sampling on the basis of unphysical phase separation. We note that the [0.205, 0.110, 0.685] feed produced unphysical phase separation for 19 out of the 20 iterations, which is why no standard deviations are reported. However, for the converging sample, we see expected patterns for both organic and aqueous phase compositions. For the remaining feeds, quick convergence into predictions with standard deviations (σ) constituting $<1\%$ of mean values across all metrics is strong evidence of the robustness of our model in predicting phase splits for a water + furfural + ethyl acetate mixture with 2.5 wt% NaCl.

To illustrate the variability implied by these metrics, Figure 9 displays mean values and standard deviations associated with the uncertainty workflow. We observe narrow ranges across all metrics, including the purity of each phase, which underscores low variability in composition and demonstrates prediction robustness through perturbations. Similarly, simulated log K values evidence low variability in the partition of furfural at equilibrium, further supporting the claim that the proposed model performs strongly despite introduced uncertainty in τ parameter determination.

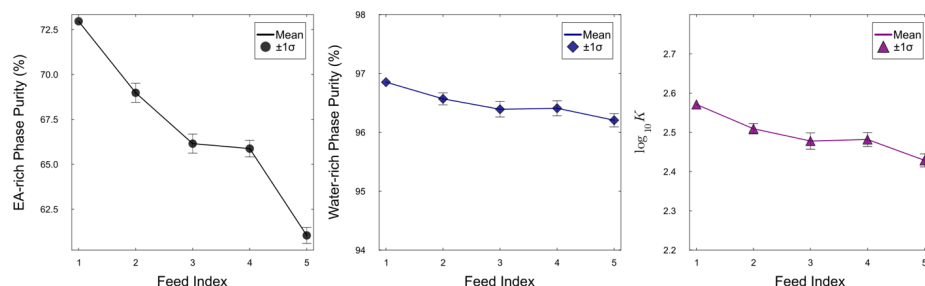


Figure 9. Phase separation metrics for different feeds following 20 perturbation instances on τ_{13} and τ_{31} , following a modified Morris method determination of the most influential τ matrix entries. Top: Average phase purity and standard deviation for the ethyl acetate-rich phase as a function of feed index (1-5 in the order reported in Table 6). Middle: Average phase purity of the water-rich phase as a function of feed index. Bottom: Average normalized furfural partition coefficient magnitude and standard deviation as a function of feed index.

3.4. Limitations and Future Work

Commented [JLG75]: I think this section was trying to do too much. Maybe separate out the parts here about analyzing your model - its limitations and future work potential. Then conclusions below should be more about the actual "what is concluded from work" and "why it matters"

Some limitations to this model arise as a direct consequence of its partially data-driven nature. The use of an optimization-based algorithm for τ search involves the minimization of phase split prediction losses across all six interaction parameters at once, which ultimately does not guarantee the thermodynamic accuracy of individual interaction parameters, particularly as sensitivity analysis results show that small perturbations to singled-out parameters only minimally alter prediction outcomes. Therefore, this workflow is not to be used for the obtention of individual thermodynamic parameters to be used in other modeling applications. An additional limitation includes the model's low extrapolative power; we observe that, when evaluating conditions significantly outside of the range covered in the training step, loss penalties for unphysical predictions are occasionally bypassed, which provides erroneous predictions. This could be resolved by expanding the range of the data represented in the training set and diversifying the starting guesses for the feed composition. However, reliable training data is still scarce in the literature due to the particularity of the system of interest, given the choice of a specific solvent and model component. In future work, we seek to test model performance against experimental data for the phase split of cellulose-derived biocrude, maintaining the current training data set that uses furfural as a model component. By collecting and testing these predictions, our aim is to evaluate whether the model component approach is practical for direct use in biocrude recovery estimations or a more nuanced training and simulation pipeline is required.

Future steps to discourage these instances will involve the development of a workflow for the refinement of guesses, which may be informed by either synthetic training data or experimental identification of feed compositions that result in one-phase regions. This workflow will also aid in the overall robustness of the model, strengthening its handling of physically inconsistent phase partitioning. Furthermore, we observed a comparable degree of predictive agreement between our proposed model and Aspen Plus V11[®] in the phase split simulation of our experimentally obtained tie lines. Similarly, solid predictions were obtained when contrasted to experimental tie lines reported in literature (de Almeida et al.), within the low-furfural and low-ethyl acetate regimes.

4. Conclusions

This work presents major advances in the predictive simulation of the salting-out separation of biocrude model components from HTL PW. Phase split predictions for the salting-out extraction of furfural in water with ethyl acetate were produced by development of a hybrid model incorporating NRTL calculations via regression of interaction parameters from synthetic (Aspen-generated) data sets. The Julia-based model environment shows high performance in τ regression, exhibiting low losses against the Aspen-generated training sets. Strong agreement was shown between the model-predicted phase separation of both synthetic and experimental data across all steps of model validation, from baseline LLE to synthetic-feed SALLE, excluding instances where the feed composition guess produces unphysical phase separation. Overall, the model demonstrates strong predictive capability across experimental outcomes for the salting-out extraction of furfural in the <10 mol% region.

Sensitivity analysis results demonstrate the robustness of the proposed model when presented with perturbations, which further supports this workflow for the prediction of non-ideal (non-synthetic) tie lines. Similarly, the transferability of the grey-box pipeline employed in this model (custom multi-step parameter regression on reference tie lines, model validation against interpolated synthetic data, and subsequent use of the regressed parameters in the NRTL

deterministic equations) makes this modeling environment highly valuable for application to other partially miscible systems beyond the salting-out extraction of biocrude-like organics.

The flexibility of this grey-box model architecture enables further training on experimental data to reliably capture non-idealities and identify physical process limitations, with potential for applications across industrial and academic SALLE implementations of varying organic solute and solvent systems. By leveraging model-based prediction capabilities, this approach aims to reduce experimental burden and resource investment in the optimization of salt-assisted separation workflows.

Author contributions

L. Capellino: Conceptualization; Methodology; Investigation; Formal Analysis; Visualization; Writing – Original Draft

S. Lin Htun: Conceptualization; Methodology; Investigation; Writing – Review & Editing

S. Parmar: Investigation; Formal Analysis; Writing – Review & Editing

J.L. Goldfarb: Conceptualization; Writing – Review & Editing; Supervision; Resources; Funding Acquisition

Conflicts of interest

There are no conflicts to declare.

Data availability

The full model discussed in this work is available on GitHub at the following link: [link].

Additional data for this article, including HPLC and GC-MS readings, are available at the corresponding author's Harvard Dataverse at URL (to be published after acceptance).

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